

An Adaptive Upper–Lower Certificate Portfolio for Directional Stein Tails

Safe Pareto Pruning, Three-Way Triage, and a Candidate Stage A1 Ledger

Liang Wang

July 2026

Abstract

RH-60–RH-68 produced several valid but incomparable tail mechanisms: finite phase fusion, geometric completion, block Krylov centers, physical-covariance PSD envelopes, weighted terminal residuals, and phase-coherence lower bounds. No single mechanism is uniformly best, and RH-68 proves that a universal fixed-depth block rule is impossible. This paper assembles the mechanisms into an adaptive certificate portfolio while preserving each input’s claim level.

A candidate consists of a valid upper U_c and a nonnegative cost vector b_c containing such entries as horizon, block depth, covariance focus, and global PSD size. We prove that Pareto-dominated candidates may be discarded without changing any monotone budget decision. The surviving portfolio has three statuses: green when an upper meets every displayed budget, red when no upper closes and a certified lower bound excludes the stated approximation class, and amber otherwise. A lower bound is never used to reject a broader class than it proves.

For an exact finite prefix F_L and any valid tail candidate,

$$\mathcal{E} \leq F_L + U_c, \quad \mathcal{E} \leq F_L + \min_{c \in \mathcal{C}_L} U_c.$$

This yields a conditional Stage A1 ledger: polylogarithmic finite terms, tail uppers, and admissible cost entries imply the required polylogarithmic Hardy bound. The physical assumptions remain unproved.

On the archived RH-60 five-scale family, a one-percent selector grows from horizon 4 to 32 and then remains at 32. At $\sigma = 0.01$, RH-61’s geometric horizons are 1111 and 307, compared with phase horizon 32. The covariance branch selects no focusing for the generic chain and complex-phase models, but selects $\varepsilon = 10^{-24}$ for exact cancellation, with physical/global gains 1.001027/489.96. Exact and jittered Fourier rings are red; wide phase arcs remain amber when only numerical misses, not analytic lower bounds, are available. A 256-bit Arb audit certifies one exact branch of each type. The architecture is now explicit, but the physical rows are still binary64 diagnostics and Stage A1 is not closed. No arithmetic trace formula or Hilbert–Polya conclusion is claimed.

Keywords: certificate portfolio; Pareto frontier; upper–lower bound; Stein tail; block Krylov method; phase-aware fusion.

MSC 2020: 47A10; 47B65; 65F35; 90C29; 93B07.

1 Introduction

The route from RH-50 to RH-68 has become a collection of local theorems rather than one monolithic estimate. This is not a defect. The physical family can be phase-compressed at one scale, nonnormal at another, and close to an exact cancellation ray at a third. RH-68 shows that selecting one fixed Krylov depth from stability data alone is mathematically impossible [3].

The natural replacement is a portfolio. Multicriteria Pareto reduction is standard [1], but a proof-oriented portfolio must also keep upper and lower claims separate. A failed upper is not a no-go theorem, and a lower bound for one projection class does not rule out finite phase fusion or a different packetization.

Contributions and boundary

1. We formalize safe dominance and Pareto pruning for rigorous upper candidates with vector-valued costs.
2. We prove a green/red/amber triage rule that does not overextend lower bounds.
3. We compose exact finite-prefix and terminal-tail certificates and give a conditional polylogarithmic budget theorem.
4. We apply the ledger to archived physical and synthetic rows from RH-60, RH-61, RH-67, and RH-68.

The portfolio logic is analytic. A row is only as rigorous as its source; binary64 physical diagnostics remain binary64 after selection.

2 Certificate candidates and safe Pareto pruning

Let $\mathcal{C}$ be a finite family of candidates. Candidate c has a nonnegative upper U_c and a cost vector

$$b_c = (b_{c,1}, \dots, b_{c,s}) \in [0, \infty)^s. \quad (1)$$

The coordinates may record horizon, block depth, covariance focus exponent, metric conditioning, global PSD gain, or interval precision. They must be defined consistently within one comparison.

Definition 2.1. Candidate c dominates d if

$$U_c \leq U_d, \text{ and } b_{c,j} \leq b_{d,j} \quad (1 \leq j \leq s), \quad (2)$$

and at least one inequality is strict. The Pareto frontier $\mathcal{P}(\mathcal{C})$ contains the undominated candidates.

Theorem 2.2 (Safe Pareto pruning). *Let $\Phi(U, b)$ be nondecreasing in every coordinate. Then*

$$\min_{c \in \mathcal{C}} \Phi(U_c, b_c) = \min_{c \in \mathcal{P}(\mathcal{C})} \Phi(U_c, b_c). \quad (3)$$

Likewise, for any upper target and coordinatewise cost budgets, a feasible candidate exists in $\mathcal{C}$ if and only if one exists on the frontier.

Proof. Every removed candidate is dominated by another candidate with no larger objective value and no larger budget entries. Repeating along the finite dominance relation reaches an undominated candidate. $\square$

The theorem is intentionally modest. It does not average incomparable certificates or convert diagnostics into validated uppers.

3 Green, red, and amber

Fix an upper target τ , cost budgets B_j , and, when available, a lower bound ℓ for a specifically named approximation error with target δ .

Definition 3.1 (Three-way triage). The ledger status is:

1. *green* if some valid upper candidate has $U_c \leq \tau$ and $b_{c,j} \leq B_j$ for every j ;
2. *red for the stated approximation class* if no candidate is green and a certified lower bound satisfies $\ell > \delta$;
3. *amber* otherwise.

Proposition 3.2 (Triage soundness). *A green status proves the displayed upper budget. A red status proves only that the class named by ℓ misses its displayed approximation target. An amber status makes neither assertion.*

Proof. Green is the definition of a valid feasible upper. Red is the contradiction between a certified lower bound and the target for the same quantity. No logical conclusion follows when neither condition holds. $\square$

This discipline matters for the phase arcs in RH-68. A binary64 depth miss is evidence, not an analytic lower bound, so wide arcs are amber rather than red.

4 Finite-prefix and tail composition

Suppose an exact finite response calculation and a positive tail mechanism give, for every admissible candidate c at horizon L ,

$$\mathcal{E} \leq F_L + U_{L,c}. \quad (4)$$

This includes the RH-60 phase-aware finite Gram followed by a geometric, block, or weighted residual completion [4].

Theorem 4.1 (Adaptive composition). *For every horizon L ,*

$$\mathcal{E} \leq F_L + \min_{c \in \mathcal{C}_L} U_{L,c}, \quad (5)$$

and the minimum may be restricted to the Pareto frontier. Taking a further minimum over a finite set of horizons remains valid.

Proof. Equation (4) holds separately for every candidate. Taking the minimum of right sides preserves the inequality; Theorem 2.2 removes only dominated entries. $\square$

Corollary 4.2 (Conditional polylogarithmic ledger). *Let $\sigma \downarrow 0$. Suppose there exist selected horizons and candidates such that*

$$F_{L_\sigma} = O(\log(1/\sigma)^a), \quad U_{L_\sigma, c_\sigma} = O(\log(1/\sigma)^b), \quad (6)$$

and every analytically relevant transfer cost in b_{c_σ} is polylogarithmic. Then the corresponding directional Hardy energy is polylogarithmically bounded and can be inserted into the conditional RH-50 bridge.

Proof. Apply (4) and add the two polylogarithmic quantities. The remaining transfer factors are covered by the stated cost assumption. $\square$

The corollary is a bookkeeping theorem, not Stage A1: none of its physical uniformity hypotheses is proved here.

5 Portfolio audit

5.1 Archived physical horizon branch

For every RH-60 row and side, candidates are the stored horizons $\{0, 1, 2, 4, 8, 16, 32, 64\}$. The upper entry is phase-aware completion divided by the archived exact Hardy energy; the cost is horizon. We select the least horizon with gain at most 1.01 and compare it with RH-61’s packetwise geometric horizon for the same tolerance.

Table 1: One-percent horizon portfolio.

σ	phase L/R	geometric L/R	saving left	saving right
0.16	4/4	8/9	2.00	2.25
0.08	8/8	22/24	2.75	3.00
0.04	16/16	67/69	4.19	4.31
0.02	32/32	220/165	6.88	5.16
0.01	32/32	1111/307	34.72	9.59

The selection is deterministic over archived binary64 values. It is not an interval-certified physical upper and does not establish an asymptotic horizon law.

5.2 Covariance and depth branches

The RH-67 covariance frontier uses physical gain as upper and (focus exponent, global gain) as costs [2]. Targets 4.0 and 1.3 select isotropic covariance for the nonnormal chain and complex-phase model. The exact cancellation target 1.01 selects $\varepsilon = 10^{-24}$, physical gain 1.001027, focus exponent 24, and global gain 489.96.

The RH-68 exact rings and all four jittered rings are red at the displayed depth budgets: their spectral lower bounds at horizon and depth 32 range from 0.97848 to 1, above the 0.1 target. For phase arcs and depth budget 8, widths 0, 0.03, 0.1, 0.3 are diagnostic green; widths 1, 3, 2π are amber, because the archive contains misses but no analytic lower bound for those particular arcs.

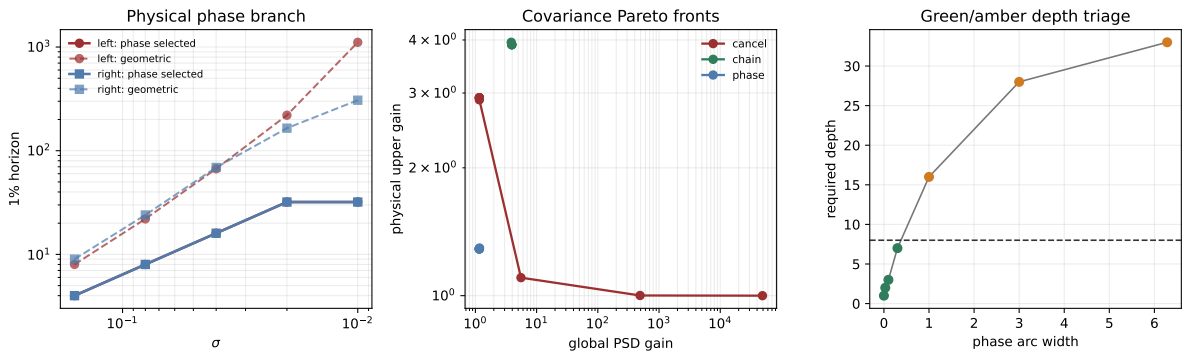


Figure 1: Left: phase fusion versus geometric horizons. Middle: covariance Pareto fronts. Right: green and amber arc-depth decisions under budget $k \leq 8$.

A 256-bit Arb audit exercises three exact branches. It certifies that $q = 0.99$ first reaches 10^{-3} at horizon 688, that the focused exact cancellation candidate has physical gain below 1.01 and global gain below 500, and that the exact Fourier lower bound 1 rejects a ten-percent projection target.

6 Route consequence

The candidate architecture is now explicit:

$$\begin{aligned} \text{finite phase fusion} &\longrightarrow \text{depth lower gate} \longrightarrow \text{block/covariance upper} \\ &\longrightarrow \text{weighted terminal residual} \longrightarrow \text{Pareto selection.} \end{aligned} \tag{7}$$

RH-70 should apply this ledger to a finite production family with outward rounding. The purpose is not yet a continuum theorem. It is to determine, without optimistic binary64 substitutions, whether each physical row is green, red for a specific route, or honestly amber. That audit will decide which asymptotic theorem is worth attempting next.

7 No arithmetic or Hilbert–Polya conclusion

This paper constructs no self-adjoint operator, no $T \log T$ counting law, no prime-power trace formula, and no completed-zeta identity. It makes no Hilbert–Polya or Riemann-hypothesis claim. Stage A1 and unconditional Stage A4 remain open.

8 Reproducibility

The directory contains portfolio logic, tests, archived-input composition, an Arb audit, figures, hashes, and publication artifacts. The main commands are:

```
pytest -q -p no:cacheprovider
python experiments/run_certificate_portfolio.py
python experiments/run_arb_portfolio_audit.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The Pareto, triage, and composition statements are analytic. Physical phase rows and generic covariance rows retain their binary64 source status. Production interval validation, a physical covariance theorem, asymptotic phase compression, Stage A1, unconditional Stage A4, a self-adjoint Hilbert–Polya operator, an arithmetic trace formula, and a zeta-zero identity remain open.

References

- [1] Matthias Ehrgott. Multicriteria optimization. *AI Magazine*, 26(3):83–94, 2005.
- [2] Liang Wang. Physical-covariance block residual envelopes. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-67-physical-covariance-block-envelopes.
- [3] Liang Wang. Phase-coherence barriers to uniform block Krylov depth. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-68-phase-coherence-block-depth-barrier.
- [4] Liang Wang. Finite-horizon phase-aware Stein tails for directional Hardy fusion. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-60-finite-horizon-phase-aware-tails.